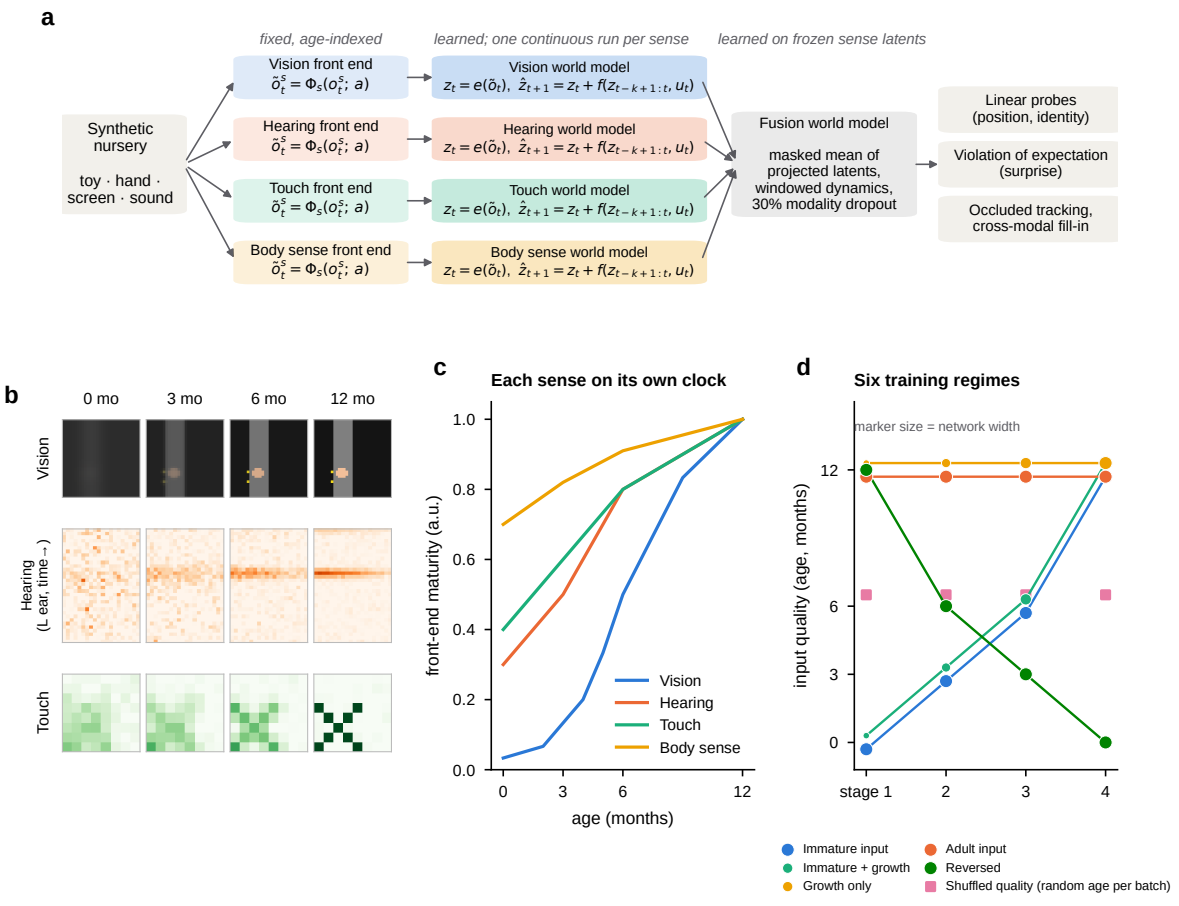

a
fixed, age-indexed
learned; one continuous run per sense
learned on frozen sense latents
Synthetic nursery
toy · hand · screen · sound
Vision front end
$\tilde{o}_t^s = \Phi_s(o_t^s;\ a)$
Hearing front end
$\tilde{o}_t^s = \Phi_s(o_t^s;\ a)$
Touch front end
$\tilde{o}_t^s = \Phi_s(o_t^s;\ a)$
Body sense front end
$\tilde{o}_t^s = \Phi_s(o_t^s;\ a)$
Vision world model
$z_t = e(\tilde{o}_t),\ \hat{z}_{t+1} = z_t + f(z_{t-k+1:t}, u_t)$
Hearing world model
$z_t = e(\tilde{o}_t),\ \hat{z}_{t+1} = z_t + f(z_{t-k+1:t}, u_t)$
Touch world model
$z_t = e(\tilde{o}_t),\ \hat{z}_{t+1} = z_t + f(z_{t-k+1:t}, u_t)$
Body sense world model
$z_t = e(\tilde{o}_t),\ \hat{z}_{t+1} = z_t + f(z_{t-k+1:t}, u_t)$
Fusion world model
masked mean of projected latents, windowed dynamics, 30% modality dropout
Linear probes (position, identity)
Violation of expectation (surprise)
Occluded tracking, cross-modal fill-in
b
0 mo
3 mo
6 mo
12 mo
Vision
Hearing (L ear, time→)
Touch
c
Each sense on its own clock
front-end maturity (a.u.)
age (months)
Vision
Hearing
Touch
Body sense
d
Six training regimes
marker size = network width
input quality (age, months)
stage 1
2
3
4
Immature input
Immature + growth
Growth only
Adult input
Reversed
Shuffled quality (random age per batch)